



SOUND-01

Class 08 - Science

Section A

1. Speed of sound in solids is ____ the speed of sound in liquids. [1]
 - a) same as
 - b) sometimes greater and sometimes less than
 - c) less than
 - d) greater than
2. A bird makes high pitched sound but a lion makes a _____ roar. [1]
 - a) High pitched
 - b) Low pitched
 - c) Without pitch
 - d) Equal pitch
3. Shrillness of sound is determined by the _____ of vibration. [1]
 - a) Height
 - b) Size
 - c) Number
 - d) Shape
4. Speed of sound _____ with increase in temperature [1]
 - a) Decrease
 - b) Remain same
 - c) Increase
 - d) May increase or decrease
5. Speed of sound _____ with increase in humidity. [1]
 - a) Do not change
 - b) Increase
 - c) Decrease
 - d) Remain same
6. The loudness above _____ becomes physically painful noise. [1]
 - a) 80 dB
 - b) 10 dB
 - c) 40 dB
 - d) 20 dB
7. Vocal cords in men are about ____ long. [1]
 - a) 2 cm
 - b) 2 mm
 - c) 20 mm
 - d) 20 cm

Section B

8. Choose the correct answer: Sound can travel through **[1]**
- a) solids, liquids and gases b) liquids only
- c) solids only d) gases only
9. Which of the following statements are correct? **[1]**

- i. Sound is produced by vibrations.
- ii. Sound requires a medium for propagation.
- iii. Light and sound both require a medium for propagation.
- iv. Sound travels slower than light.

a) ii, iii & iv only

b) i & ii only

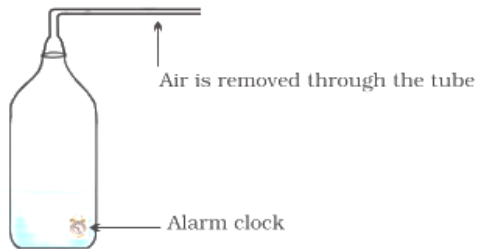
c) i, ii & iv only

d) i, ii & iii only

Section C

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| 10. | Is sound is produced by all bodies. | [1] |
| 11. | What is the loudness of a normal conversation? | [1] |
| 12. | Name a musical instrument which produces sound by air column. | [1] |
| 13. | What is called musical sound? | [1] |
| 14. | How loud sound is harmful to us? | [1] |
| 15. | Does any part of our body vibrate when we speak? Name the part. | [1] |
| 16. | What are vibrating parts of Flute? | [1] |
| 17. | What frequencies are heard and produced by dogs and bats respectively? | [1] |
| 18. | What sources in home may lead to noise? | [1] |
| 19. | What is sound? | [1] |
| 20. | What type of voice is produced when the vocal cards are tight & thin? | [1] |
| 21. | What sources in the home may lead to noise? | [1] |
| 22. | Write a common ill effect of noise pollution. | [1] |
| 23. | What is noise pollution? | [1] |
| 24. | What is the pitch of the sound? | [1] |
| 25. | On which principle, the Indian musical instrument "Jal Tarang" works? | [1] |
| 26. | What is the range of audible frequencies for a normal human ear? | [1] |
| 27. | Define vocal cords. | [1] |
| 28. | What is the unit of loudness? | [1] |
| 29. | Explain the importance of sound in our daily life. | [1] |
| 30. | What do you feel when you touch a sound producing school bell? | [1] |
| 31. | When we hear a sound, does any part of our body vibrate? Name the part. | [1] |
| 32. | How is sound produced? | [1] |
| 33. | What do you understand by vibrations? | [1] |
| 34. | In some cases we cannot see vibrations explain why is it so? | [1] |
| 35. | What is an echo? | [1] |
| 36. | Give examples of noise and musical sound. | [1] |
| 37. | Name two musical instruments which produce sound by vibrating strings? | [1] |
| 38. | What type of voice is produced when the vocal cards are tight and thin? | [1] |
| 39. | What is amplitude? | [1] |
| 40. | What is the normal length of vocal cords in man? | [1] |
| 41. | What do you understand by stringed instruments? | [1] |
| 42. | What do you mean by air pollution? | [1] |
| 43. | What do you mean by the term loudness? | [1] |
| 44. | What do you mean by an oscillatory motion? | [1] |

45. What is velocity of sound? [1]
46. Write any one method to control noise pollution. [1]
47. Touch the bell when it stops producing sound. Can you feel the vibrations? [1]
48. What are radio waves? [1]
49. An alarm bell is kept inside a vessel as shown in Fig. A person standing close to it can distinctly hear the sound of the alarm. Now if the air inside the vessel is removed completely how will the loudness of alarm get affected for the same person? [3]



50. Write any five sources of noise in your surroundings. [3]
51. What is an Echo? When one can hear the echo of sharp sound? Why cannot we hear an echo in a small hall? [3]
52. a. Which wave property determines: [3]
- i. loudness
 - ii. pitch
- b. What is Sound?
53. a. What is one vibration in a second called as? [3]
- b. What is oscillation? A vibrating object produces 156 waves in four second. Calculate the frequency of the vibrating object.
54. Define frequency, amplitude and time period of a vibration. [3]
55. What is ultrasound and where it is used? [3]
56. What is frequency range for human ear? What is audible and inaudible sounds? [3]
57. A pendulum oscillates 40 times in 4 seconds. Find its time period and frequency. [3]
58. The town hall building is situated close to Boojho's house. There is a clock on the top of the town hall building which rings the bell every hour. Boojho has noticed that the sound of the clock appears to be much clearer at night. Explain. [3]
59. How do bats catch their prey? [3]
60. What is difference between noise and music? [3]
61. Lightning and thunder are produced simultaneously. But thunder could be heard a few second after lightning, why? [4]
62. Show experimentally that sound waves can travel through liquids. [4]
63. How do trees planted along roadside help in the reduction of noise pollution? [4]
64. What is speed of sound? A child receives his echo 4 sec later. Find the distance of the reflecting surface from the child if speed of sound in air is 342 m/s. [4]
65. Sound needs a material medium to propagate. Explain? [4]
66. Write the applications of the ultrasound. [4]
67. How does human produce sound? [4]
68. Explain how the human ear works? [4]
69. What is eardrum? How does it is play an important role in hearing? [4]
70. What do you mean by loudness and pitch of sound and how they are affected? [4]

71. How is sound produced and how is it transmitted and heard by us? [4]
72. Draw a diagram: [4]
- a. Soft sound and a louder sound.
 - b. High pitch and low pitch.
73. What is noise pollution? How does it affect us and how it can be prevented? [4]
74. What is the difference between percussion and string instrument? [4]
75. What would happen if there were no voice boxes in our throat? [4]